

CSE 3252: Web Engineering Lab

Credits	1
Contact Hours	28
Year	Third
Semester	Second
Prerequisite	CSE 2232: User Interface Development Lab
Course Type	☐ Theory ☒ Laboratory work ☐ Project work ☐ Viva Voce

Motivation

Web Engineering Lab complements the theory course by giving students hands-on experience in building full-stack web applications. Using modern frontend and backend frameworks, students will learn how to implement secure, database-driven, and maintainable web systems.

Course Objective

This laboratory course develops practical skills in full-stack web development using React on the frontend and a modern server-side web framework on the backend. Students will design APIs, connect databases, implement authentication, integrate frontend and backend modules, test web applications, and package them using modern development tools.

Course Outcomes (COs), Program Outcomes (POs) and Assessment

CO No.	CO Statement	Corresponding PO	Domain / Level of Learning Taxonomy	Delivery Methods & Activities	Assessment Tools
CO1	To construct RESTful backend services and database-driven features using a modern server-side web framework	Modern Tool Usage (PO5)	Psychomotor domain – Level 4	☐ Lecture Note ☒ Text Book ☐ Audio/Video ☒ Web Material ☒ Lab Manual	☒ CA ☒ Final Exam ☐ Assignment ☐ Notebook ☒ Presentation
CO2	To integrate React frontends with backend APIs, authentication, validation, and persistence to develop full-stack web applications	Design/Development of Solutions (PO3)	Cognitive domain – Level 6	☐ Lecture Note ☒ Text Book ☐ Audio/Video ☒ Web Material ☒ Lab Manual	☒ CA ☒ Final Exam ☐ Assignment ☐ Notebook ☒ Presentation
CO3	To use Postman, Git, and Docker to test, version, and containerize secure web applications	Modern Tool Usage (PO5)	Psychomotor domain – Level 4	☐ Lecture Note ☒ Text Book ☐ Audio/Video ☒ Web Material ☒ Lab Manual	☒ CA ☒ Final Exam ☐ Assignment ☐ Notebook ☒ Presentation

Assessment and Marks Distribution

- Continuous Assessments (CA) (30%)
- A comprehensive final exam + Lab note book (60%)
- A class participation mark (10%).

Lab Course Contents / List of Experiments

- Lab 1 — Environment Setup and Project Structure
 - Lab 2 — Backend Framework Basics and Database Migrations
 - Lab 3 — REST API Development with Postman
 - Lab 4 — Authentication and Role-Based Access
 - Lab 5 — React-API Integration
 - Lab 6 — Full-Stack CRUD Module
 - Lab 7 — File Uploads, Security, and Validation
 - Lab 8 — Testing, Debugging, and API Documentation
 - Lab 9 — Dockerization and Capstone Project
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Textbooks

1. Matt Stauffer — *Laravel: Up & Running* (3rd ed.), O'Reilly Media, 2023
2. Ethan Brown — *Web Development with Node and Express* (2nd ed.), O'Reilly Media, 2019